

Biases and Fairness in Deep Learning Models: A survey on inculcating fairness in generative models

Grishma Deshmukh
Cummins College of Engineering for Women
Pune, India
grishma.deshmukh@cumminscollege.in

Abstract— AI-generated images are now widely used in various industries, including the art industry, advertising, healthcare, etc. The generative models used in making this happen are only getting incrementally powerful. However, as with any technology, these models are not immune to the inherent biases which are inadvertently introduced due to the discrimination reflected in the data. These biases in data can lead to undesirable consequences, such as reinforcing stereotypes, discrimination or perpetuating inequalities. Hence 'Fairness' in generative models is an increasingly crucial concept. This review paper discusses the biases and their types that percolate the deep learning models and are reflected in their results. The paper then presents a comprehensive overview of FairGAN and FairGAN+, two recently proposed generative models that aim to generate fair, accurate and diverse data, including their strengths and limitations. FairGAN incorporates fairness constraints into the objective function to generate fair data. FairGAN+ improves upon FairGAN by using multiple discriminators to achieve both fairness and accuracy in the generated data while also promoting diversity. Overall, FairGAN and FairGAN+ are promising approaches to addressing fairness in generative models. Further developments in this area are expected to lead to more effective techniques for generating fair and accurate data.

Keywords—Artificial Intelligence, Deep Learning, Algorithmic Bias, Fairness, Generative Models, Generative Adversarial Networks, Auxiliary Classifier GAN, Adversarial debiasing, FairGAN, FairGAN+

I. INTRODUCTION

With the burgeoning development of deep learning models, generative networks capable of producing images and artworks are becoming more and more realistic. Their capabilities expand from generating images from a text prompt to writing a solution code for a problem statement. Many generative algorithms and models have been developed to imitate a brush stroke or an artist's painting style, using artificial intelligence and deep learning methodologies such as Variational Autoencoders, Generative Adversarial Networks, Transformers, etc. These models use data available from across the internet. The data being directly and indirectly generated by humans, is subject to a lot of ethical bias and unfairness. Algorithmic bias occurs when the algorithm produces outputs that are prejudiced as a result of the bias in the data or amplification of that bias due to false assumptions during the training process. Fairness refers to the attempts to mitigate or get rid of the algorithmic bias to ensure the absence of partiality and prejudice against a set of data based on its inherent features.

There are numerous examples of incidents where a model has explicitly exhibited discrimination in its outcomes. The software Correctional Offender Management Profiling for Alternative Sanctions (COMPAS) is one such representative

case, as detailed in [1]. The application is used to gauge the risk of an individual to recommit another crime. However, when the model's outcomes were evaluated, it was found to have a biased disposition against African-Americans. Upon investigation, the software appeared to have a higher tendency to produce biased decisions when it came to predicting in the cases of offenders with African-American profiles compared to the ones with Caucasian profiles. Another example can be seen in the form of gender bias where the Artificial intelligence (AI)- generated images generated more images of men than women for a text prompt expecting results for a person in STEM (Science, technology, engineering, and mathematics).

As generative technology advances, the concern regarding the spread of false information and synthetic media like deep-fakes is ever-increasing. Efforts to mitigate or eliminate the bias and traces of discrimination are being constantly made, now more than ever.

A. Discrimination, Bias and Fairness Models

Despite the continuous development and improvisations in the accuracy of generative models, the issues concerning generative models are the biased predictions that stem from the latent prejudice and its perpetuation in the model's output. Discrimination is the source of unfair representation and bias. It can be the result of intentional or unintentional stereotyping of some sensitive attributes. The discrimination can manifest itself in the outcomes of a deep learning model and might even be amplified. There are two main sources of discrimination which are Systematic discrimination and Statistical discrimination. Systematic discrimination occurs due to the customs, behaviour aspects, or policies in an organization. Statistical discrimination occurs when average group statistics are used to evaluate a subgroup or individual without any special attention to the inherently biased attributes.

Bias refers to the "privileging" of one category over another. Bias is the manifestation of discrimination present in data. An example of a bias can be that a generative model is more likely to produce images influenced by American culture than Asian/ African cultures unless specifically mentioned otherwise. The paper [2] discusses the categories of biases and the types under them. They can be summarized as:

- Data to Algorithm bias: Bias in data that result in biased outcomes. For example, due to the majority of records containing information about men as compared to women, a model results in a higher possibility of decisions pertaining to men than women. The types are- Sampling Bias, Measurement bias, Longitudinal Data Fallacy,

Representation bias, Aggregation bias, Linking bias, Omitted Variable Bias, etc.

- Algorithm to user bias: The bias that occurs as a result of user behaviour that are consequences of the algorithm's modulation. For example, if a series is placed in the first row or at an easily visible portion of the screen in a user interface of a streaming platform, that show is more likely to have higher viewers despite not aligning with the individual's interests. The types are: User interaction bias, Algorithmic bias, Emergent bias, Popularity bias, Evaluation bias, etc.
- User to Data bias: This type of bias is the result of human or users' bias reflected in the human generated data. For example, the data generated by first-world countries is higher than some of the smaller third-world countries due to limited access. This results in a poor representation of the former which can engender a bias. The types are: Social bias, Historical bias, Temporal bias, Population bias, Self-selection bias, Behavioural bias, Content production bias, etc.

Various algorithms and methodologies have been developed to counteract these biases. They can be implemented at any phase of the whole process. We can categorize these methods into three categories which target the mitigation or removal of biases:

- Pre-processing: Applying pre-processing techniques to modify the dataset in order to incorporate fairness and decrease bias in the data. For example- Community detection, Word embedding, Preferential sampling, etc.
- In-processing: The deep learning models change their learning and training process by imposing a constraint or modifying their objective function. Some tweaks or regularizers are applied to the model during its training phase. For example- fair classification, fair regression, etc.
- Post-processing: This is followed after the training process by applying a threshold while treating the model as a black box. These methods directly modify the outcomes of the models. For example- post-processing algorithms to achieve fairness as proposed in paper [2].

Algorithmic fairness refers to the idea that machine learning models should not discriminate against any group of people or individuals based on gender, race, religion, etc. It is becoming increasingly important as more and more decision-making processes are being automated. It is of extreme importance in fields of criminal justice, employment, finance, etc. where biased decisions have a significant social or economic impact on people's lives.

It has been defined in many ways in different papers [2], [6], [7], [8], [9] as- Equalized Odds, Equal Opportunity, Demographic Parity, Conditional statistical parity, Fairness in relational domains, Counterfactual fairness, etc. Consider a model having a classifier η and a labeled dataset D . Given that this dataset consists of a protected attribute S having binary values and a set of unprotected attributes X . The

classifier classifies the samples into two classes. Hence, D includes a binary class label Y . Considering such a case, the commonly used metrics for measuring and evaluating fairness can be defined as:

- Demographic parity: It is a measure of fairness that ensures that the proportion of positive outcomes is equal across different demographic groups. In other words, it ensures that the model does not discriminate against any particular group based on their demographic characteristics such as race, gender, or age. It is in [7] defined where, $P(\eta(X) = 1|S = 1)$ is equivalent to $P(\eta(X) = 1|S = 0)$. Demographic parity ratio (DPR) is one of the measures used to measure model bias.
- Equality of odds: It is defined as another criterion for evaluating fairness in machine learning models. Equality of odds means that the probability of a true positive prediction and that of a false positive prediction should be equal for all groups in the population, regardless of their demographic characteristics. According to [7], it is defined when for $y \in \{0, 1\}$, the value of $P(\eta(X) = 1|Y = y, S = 1)$ equals to that of $P(\eta(X) = 1|Y = y, S = 0)$. If equality of odds is achieved, it means that the machine learning model does not only treat all individuals fairly but also makes the same type of errors at the same rate across all groups.
- Equality of opportunity: It is similar to Equality of Odds but is actually its special case. Equality of opportunity means that the probability of a true positive prediction should be equal for all groups in the population, regardless of their demographic characteristics. In other words, equality of opportunity requires that the probability of a true positive prediction is the same for all groups in the population, given that the true label is positive. If equality of opportunity is achieved, it means that the machine learning model does not discriminate against any particular group when predicting the positive class and gives everyone an equal chance to benefit from the positive outcome.

These concepts provide a structured idea of the inequality in the likelihood of false positives or false negatives. Generative models create synthetic data to resemble the real data distribution. However, the incurred discrimination in the real data can be reflected in the synthetic data whenever the degree of similarity is high. Efforts to ensure fairness in deep learning models are being taken constantly. Models are constantly under development to improve their results and towards unbiasing the data. Fair machine learning methods like Fair classification, Fair regression, Fair PCS, Structured prediction, Community detection, etc. Deep Learning models have also taken steps in the direction of Fair Representation learning, Fair natural language processing, etc. Inculcating fairness during in-process or the training phase helps in focusing on optimizing the objective function of every group while also taking into consideration their sensitivity, in contrast to that of the whole dataset without any attention to the unprotected groups.

Fair classification is a method aimed for classifiers to achieve equal treatment and prediction quality amongst

sensitive attributes. It can be achieved through various methods such as:

- Counterfactual fairness which first identifies a set of counterfactual examples or scenarios that would result in a fairer outcome, and then modifies the training data to reflect these counterfactuals.
- Adversarial training which involves training the model to withstand manipulation of the model towards a particular bias or discrimination.
- Regularization involves adding regularized terms to the model's loss function, which penalizes the model for making decisions that are unfair or discriminatory towards certain groups.

Different approaches are used to implement fair classification like imposing fairness constraints into a Multitask learning framework, modified Naive Bayesian classification, etc. More research is being done to understand and combat the preferential outcomes by using the concepts of equity over equality, searching for unfairness, and synthesizing a single definition of fairness.

II. METHODOLOGIES

In this section, we will review the intersection of generative technology and fairness in machine learning. First, we will provide an overview of generative models and focus on the widely popular Generative Adversarial Networks (GANs). Next, we will delve into the importance of fairness in generative models and explore the challenges that arise when generating fair and unbiased data. Finally, we will introduce two recent advancements in this area: FairGAN and FairGAN+. These models aim to mitigate fairness issues in generative models by incorporating fairness constraints during training. We will discuss the architecture and performance of these models and their potential impact on promoting fairness in machine learning applications.

A. Generative Technology

1) *Generative Models*: Generative models are a section of artificial intelligence implementations that are used for creating more data from existing data. A generative model provides specifications behind data creation. They are used to produce fresh data, hence addressing a crucial issue of dataset creation in the field of data science. Every sample in the training dataset is called an observation, made up of various features. A generative model trains on these observations and generates a new set of data consisting of features that appear to abide by the same rules as the original dataset. The model should be able to train to approximate the input patterns and produce outputs that are different and unique from the original set of observations. They analyse patterns and relationships within the training dataset. There are various generative model architectures such as diffusion models, variational autoencoders (VAEs), Transformers, Flow-based models, Generative Adversarial Networks (GANs), etc. Each of these models uses a different approach to generate new data but they all rely on some form of statistical analysis and pattern recognition. Generative models also have many applications such as generating visual media, natural language text, simulating complex systems and processes and even generating music.

They are very frequently used to create what is widely termed as 'AI-generated images'.

One of the main challenges with generative models is ensuring that the synthetic data generated is of high quality for the intended application. This can be difficult as the model must balance the need for novelty with the need for accuracy and coherence.

2) *Generative Adversarial Networks*: They are a class of profound machine learning techniques that have received a lot of notable admiration lately because of their capacity to create new, reasonable information that intently resembles the original data. They were first introduced in 2014. They were proposed by Ian Goodfellow et. al. in [3]. The core concept behind a GAN is to make two neural networks compete against one another in a situation akin to a game, where success is defined by the first neural network's ability to fool the second one.

GANs are generative neural networks, which use an unsupervised learning approach, and aim to create realistic data from noise and some conditioning variables (such as class labels). Their architectures usually consist of multilayer neural networks. A generator exists whose function is to synthesize fake data samples using the original data. The number and main functionality of the discriminators may vary in different models. However, in most proposed architectures, the function of the primary discriminator is to classify and predict whether the input is the actual data from the original dataset or synthesized data, produced by the generator.

As the generator gets advances at producing data that looks realistic compared to the real dataset, the discriminator gets smarter at recognizing counterfeit data. Eventually, the generator can make engineered data essentially virtually indistinguishable from the original. The successfulness of the GAN is determined by the generator's ability to delude the discriminator into mistaking fake data for real. This indicates that the data created is a close approximation of data that can be produced by the original set of rules.

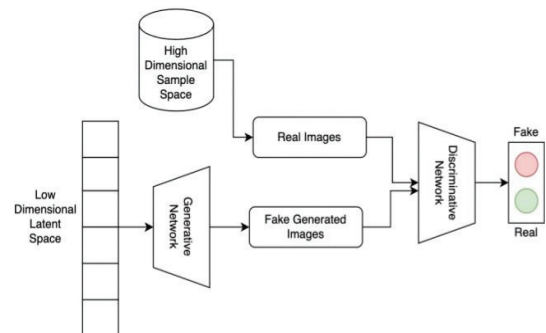


Fig. 1. Generative Adversarial Networks Architecture

The few most popular GAN models are CycleGAN, Vanilla GAN, StyleGAN, DiscoGAN. The most widely used applications are to transform an image to a different domain (for example, to transform an image to an artist's art style), generate an image from a textual prompt, generate high-resolution images, etc. They are implemented in various ways from creating realistic images and videos to generating synthetic medical data for research purposes. They are also used in creative fields such as art and music,

where they are used to generate new, unique works that are inspired by existing pieces. Despite their success, GANs still face numerous challenges such as the risk of over-fitting.

B. Fairness in Generative Models

Generative models have made remarkable strides in the past few years, enabling the synthesis of realistic images, sounds, and text. However, they have also been found to reflect and enhance the underlying predispositions in their training data, perpetuating inequalities, and discrimination against certain groups. To solve this issue, researchers have devised a number of strategies for improving the fairness of generative models. This section will discuss some of the most prominent approaches, including Adversarial Debiasing (AD), Auxiliary Classifier GAN (ACGAN), FairGAN, and FairGAN+, highlighting their strengths and limitations in promoting fairness and combating bias.

1) *Adversarial Debiasing (AD)*: Adversarial debiasing, proposed in the paper [4], is an in-processing technique used in machine learning to mitigate bias in models and improve their fairness. It involves training a model on a dataset that has been deliberately modified to introduce bias, in order to make the model more robust and less susceptible to such biases. To increment the accuracy of predictions, it trains a classifier. Doing so also helps in declining an adversary's capability to infer the protected or sensitive attribute from the predicted outputs.

The methodology includes two vital parts: a classifier that predicts the target variable of interest, and an adversarial network that tries to predict the sensitive or the protected attribute (e.g., race, gender) with the features utilized by the classifier as the basis. For the classifier to produce accurate predictions free from bias, the adversarial network must learn an uninformative representation of the data for the protected attribute. During training, the classifier and adversarial network contend with one another in an adversarial game of minimax, where the objective of the classifier is to maximize prediction accuracy while the objective of the adversarial network is to maximize the error rate in estimating the protected attribute. The resulting model is expected to be fairer and less biased in making predictions about individuals from different demographic groups.

For example, suppose we want to train a facial recognition system using a dataset of images. However, we observe that the dataset is biased towards lighter-skinned individuals, which can lead to inaccurate or unfair predictions for people with darker skin tones. To address this bias, we use adversarial debiasing. We train the two neural networks simultaneously: the classifier network that receives an input source image and predicts the identity of the person in the image, and the adversarial network that also receives the same image as input, attempts to remove any information in the image that the classifier might use to make biased predictions, such as skin colour. The adversarial network returns a modified image to the classifier network for prediction. After training, the classifier network should be able to accurately predict the identity of a person regardless of their skin colour. The adversarial network has learned to remove the correlation between skin colour and identity, which helps reduce bias in the predictions made by the classifier.

Because the predictions cannot contain any group-discriminating information that the adversary could use against them, this technique produces a fair classifier. Adversarial debiasing has been used to lessen the influence of biases in decision-making systems in numerous applications, such as computer vision, medicine, criminal justice loan approvals, natural language processing, etc.

2) *Auxiliary Classifier GAN (ACGAN)*: The Auxiliary Classifier GAN (ACGAN), proposed in the paper [5], is a varied adaptation of the generative adversarial network architecture that allows the conditional generation of images. ACGAN builds on the traditional GAN architecture by introducing an auxiliary classifier that helps to guide the generation process. This auxiliary classifier supports controlling the generator network's output on a specified condition. In an ACGAN, for a random noise vector as the input, the generator network attempts to produce an output that is similar to the original training dataset. However, in addition to generating realistic images, the generator also produces a class label that describes the type of image it has generated. The discriminator network in an ACGAN has two main components: a regular binary discriminator which works along with an auxiliary classifier that tries to classify the generated images into different categories. The auxiliary classifier takes as input the output of the generator, which includes both the generated image and the class label produced by the generator. The discriminator is trained to predict both the binary output (real or fake) and the class label of the generated images. The expansion of the auxiliary classifier in ACGAN considers more command over the generated images and assists with guaranteeing that the synthesized data is comparable to the original data in the corresponding class. By controlling the class label input to the generator, users can generate images that belong to a specific category or have certain attributes.

For example, if the training dataset consists of images of different types of flowers, the generator can be trained to generate images of specific types of flowers by inputting the corresponding class label. ACGAN has been effectively implemented in making diverse and controllable high-quality images. Although, it may not be specifically designed for reducing bias in datasets. However, ACGAN has been used in various applications like text-to-image synthesis, style transfer, anomaly detection, image synthesis, and editing.

3) *FairGAN*: The FairGAN is a type of GAN architecture, proposed in paper [6], that is designed to generate images while mitigating the effect of bias in the data by introducing fairness constraints into the training process. It was proposed as a solution to address the issue of fairness in GANs, which can inadvertently perpetuate or amplify existing biases in the training data. FairGAN puts to use a fairness loss function to penalize the network for producing biased outputs, which can help to mitigate bias in the resulting images.

The FairGAN architecture includes two principle parts. The first part is the generator which is fed a random noise vector. Its core function is to synthesize fake data that is in

close resemblance to the original dataset. The second part consists of the discriminators. The function of this part of the architecture is to differentiate between the synthesized generator-produced data and real data. Additionally, it predicts a sensitive attribute of the input image, such as race or gender. The conventional GAN loss function and a fairness loss function are two loss functions that the FairGAN model optimises through training. In order to lessen the impact of data bias, the fairness loss function is made to minimise the correlation between the predicted sensitive attribute and the output image. By minimizing the correlation amongst the predicted sensitive attribute and the output image, the FairGAN model aims to generate images that are not biased towards a particular group. Overall, the goal of FairGAN is to generate images that are not biased towards a particular group and to promote fairness and inclusivity in the generation process. It has been shown to be effective in mitigating the effect of bias in image generation and could have important applications in fields where fairness and inclusivity are critical, such as healthcare, law enforcement, and hiring. However, FairGAN may be less efficient than other methods as it requires additional computations to enforce the fairness constraints.

4) *FairGAN+*: The FairGAN+ is an improved version and an extension of FairGAN, proposed in paper [7], a variant of generative adversarial network (GAN) that's intended towards mitigation of the effect of bias in the data during its generation. Like FairGAN, FairGAN+ aims to generate images that are not biased towards any one group and promote fairness and inclusivity in the generation process. The main difference between FairGAN+ and FairGAN is that FairGAN+ introduces an additional loss function during training. FairGAN+ introduces a third loss function that is intended to promote diversity in the generated images. In addition to the standard GAN function to calculate loss and the fairness loss function used in FairGAN, the newly introduced diversity loss function determines the disparity in diversity between the generated and real data. By encouraging the generated images to be diverse, FairGAN+ aims to ensure that the model-produced data represent a more extensive scope of individuals and avoid generating stereotypical images. Overall, the goal of FairGAN+ is to generate images that are not only fair but also diverse and to promote both fairness and inclusivity in the generation process. It has been shown to be effective in mitigating the effect of bias in image generation and promoting diversity in the generated images.

To summarize, FairGAN+ has been shown to outperform the other methods in terms of accuracy and fairness measures in certain datasets and scenarios. ACGAN has been used successfully in image classification tasks and has been shown to achieve good performance while promoting fairness. FairGAN uses a minimax game between two neural networks to generate fair synthetic data but may suffer from mode collapse, which can limit the diversity of the generated data. Adversarial debiasing is a popular method for mitigating bias in classification tasks but has been shown to have limited success in certain scenarios.

C. FairGAN and FairGAN+

The implementation of Generative Adversarial Networks (GANs) to improve fairness, despite using a dataset with conspicuous bias, is being researched and improvisations are proposed. The FairGAN and FairGAN+ models are two notable approaches that have been proposed to resolve the issue of bias mitigation in data generation to ensure fair representation. In this section, we will be focusing on these two models.

1) *FairGAN Architecture*: An approach towards ensuring fairness in adversarial learning is the Fairness-aware generative adversarial network-FairGAN proposed in the paper [6]. The FairGAN model ensures achieving fair classification along with fair data production. It aims to eliminate the disparity in treatment as well as in impact from both categorical and numerical samples while preserving good data utility i.e., the synthesized data should preserve the relationships underlying amongst the features and the decision that exist in the original dataset. Its goal is to equalize the protected groups statistically by achieving statistical parity. In the implementation of FairGAN, an effort to achieve fairness, the concept of Fairness in labelled dataset and in a classifier is used.

The model's architectural design, as proposed in [6], consists of a generator G_{Dec} and two discriminators D_1, D_2 as shown in Figure 2, which has been adapted from the work of [6]. The FairGAN first generates a fake dataset conditioned on the protected attribute. For a dataset D consisting of variables: X is the unprotected attribute, Y denotes a binary decision and S is a binary value denoting the protected attribute. The original dataset, as we know, is comprised of bias and hence to ensure good quality representation of all groups and unbiased decision-making, there is a need to inculcate fairness. The aim of this model is to generate a fair dataset $\tilde{D}$, correspondingly composed of $\{\tilde{X}, \tilde{Y}, \tilde{S}\}$, which is free from disparity in treatment (fairness in data generation) and impact (fairness in classification).

The Generator produces fake samples of continuous and discrete data using the protected attribute and in accordance with a conditional distribution, it creates a fake pair that is subjected to a noise variable. The two discriminators work on input from the Generator and the original data where $D1$ ensures that the generated data is a near good representation of the original, true dataset. Followingly, a fairness constraint is applied to $D2$ which distinguishes whether the sample is from a protected or an unprotected group. The second discriminator makes certain of the absence of any correlations connecting the unprotected attribute, protected attribute, as well as with the class label. The primary goal of this model is fairness, which is accomplished by introducing a constraint to the produced data that guarantees statistical parity/fairness in the labelled dataset. The model first uses the original dataset containing indications of discrimination to pre-train an autoencoder. The decoder can then generate samples from the generator's representation as a function of the protected attribute and a noise variable. It is fine-tuned to optimize and eliminate the discriminative information. The previous models like NaiveGAN could only achieve fairness in the data generation part but not in classification.

The FairGAN has numerous applications in various fields to mitigate the bias issue and additionally, it also

helps engineer synthetic data in domains where collecting large-scale, diverse data might be considered unethical.

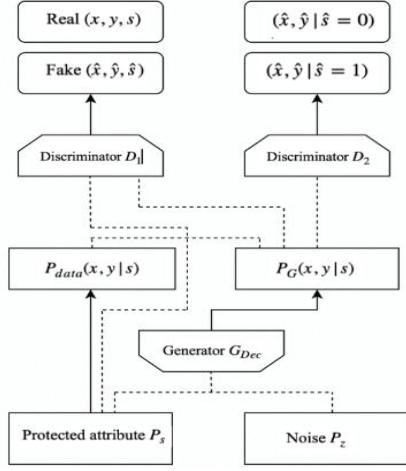


Fig. 2. FairGAN Architecture

2) *FairGAN+ Architecture*: Using inspiration from the Auxiliary classifier generative adversarial network, FairGAN+, proposed in [7], is an improved version of FairGAN (ACGAN) which in addition to producing accurate fair data, uses a classifier that generates unbiased predictions. As we can observe from Figure 3., as reproduced from the source [7], in FairGAN+, there are three discriminators and they work together with the generator G and classifier η to ensure minimal bias in data generation and fairness in classification. The first discriminator D_1 acts as the primary discriminator. The second discriminator D_2 is the fairness discriminator which ensures that the generated data is fair and does not perpetuate biases or discrimination against certain groups. The extended third discriminator D_3 , is responsible for ensuring diversity in the generation of data. It acts as the diversity discriminator. Its function is to measure the variety of the generated information by assessing the dispersion of the produced data across various locales of the input space.

Like its predecessor, with a noise variable as its input, FairGAN+ also starts with the generator training to synthesize fair data samples which follows a conditional distribution. In addition, this is followed by the inclusion of a classifier η which predicts the class label Y . Hence, to ensure proper representation of underlying data distribution, it is necessary that this synthesized data is accurate enough. The function of the first two discriminators D_1 and D_2 is similar to those in FairGAN. The addition of the third discriminator D_3 , was to differentiate the protected or sensitive attribute's values from the classifier's predictions, i.e., to determine whether the protected attribute is the basis for the projected label. This discriminator not only ensures that the classifier has good classification utility but also makes sure that the fairness metrics such as equality of odds or demographic parity are of desirable values. Like in every GAN architecture, FairGAN+ also involves joint adversarial games in its training phase. By producing artificial data that closely resembles the original actual data, the generator attempts to deceive. By attempting to deceive so that the

data created satisfies fairness conceptions in data creation and deceiving so that its predictions satisfy fairness conceptions in classification, it also engages in an adversarial game. Although it functions as a fairness restriction for data generation, classification also functions in a similar manner.

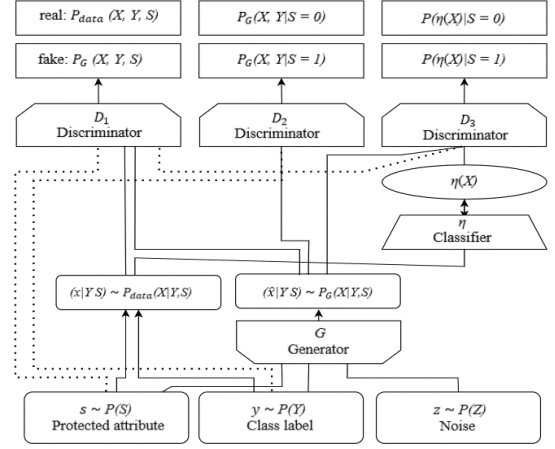


Fig. 3. FairGAN+ Architecture

D. FairGAN versus FairGAN+

Despite being better than its naive counterparts, FairGAN has its own limitations. The fact that classification-based fairness ideals like equality of odds or of opportunity can't be attained solely by producing fair data is a significant flaw in FairGAN. FairGAN is also limited in its ability to handle imbalanced data. When there is a significant disparity in the number of samples for different classes, FairGAN may produce biased or inaccurate results. FairGAN+ addresses this limitation by incorporating a class-balancing technique that ensures that each class has an equal representation in the training data. The inability of FairGAN to be interpreted is another drawback. This means that it may be challenging to comprehend how the model makes decisions or to spot any potential sources of bias. FairGAN+ addresses this limitation by incorporating a feature attribution technique that allows users to identify which features are most important for generating specific outputs. In addition, FairGAN is limited in its ability to handle missing data and may produce biased or inaccurate results in case of missing values. FairGAN+ addresses this limitation by incorporating a data imputation technique that fills in missing values with estimated values based on the available data. Moreover, models working with the FairGAN-generated fair data cannot assure fair classification.

III. RESULT ANALYSIS

FairGAN+ can attain both fairness in data generation and classification with good classification utility compared to other methods such as ACGAN and adversarial debiasing as well as FairGAN. The comparative study is based on the UCI Adult Dataset, which is a popular dataset

TABLE I. FAIR DATA GENERATION AND CLASSIFICATION ANALYSIS

Fair Data Generation						
	Metrics	UCI Adult Dataset	ACGAN	FairGAN	FairGAN+ based on Demographic Parity	FairGAN+ based on Equality of odds
Fairness	Risk Difference	0.1989	0.0120±0.0088	0.0411±0.0295	0.0106±0.0081	0.0116±0.0087
	Balanced Error Rates	0.1538	0.1964±0.0033	0.3862±0.0036	0.3867±0.0049	0.3207±0.0121
Fair Classification						
	Metrics	Logistic Regression	ACGAN	Adversarial Debiasing	FairGAN	FairGAN+
Demographic Parity (DP)	Risk Difference	0.1834	0.1119±0.0182	0.0760±0.0058	0.0901±0.0220	0.0141±0.0065
Equality of odds (EO)	Difference in True Positive Rates	0.1017	0.0854±0.0316	0.0388±0.0234	0.1473±0.0608	0.0312±0.0316
	Difference in False Positive Rates	0.0746	0.0395±0.0098	0.0184±0.0121	0.0361±0.0145	0.0245±0.0124
Classification Accuracy		0.8448	0.8359±0.0017	0.7902±0.0043	0.8256±0.0021	0.8178±0.0035 (DP) 0.8218±0.0062 (EO)

^a. This table is based on results provided in the paper [7].

generally used for classification tasks. It comprises of various fields like age, education, gender, race, income, occupation, etc. The dataset comprises a total of 48,842 samples. For the comparative analysis in [7], the income field is considered as the class label. While testing for fairness, the gender (male/female) is considered as the protected attribute. In Table I, we compare the fair data generation between the real dataset with ACGAN, FairGAN, FairGAN+. The fair classification is evaluated between a logistic regression, a fairly simple statistical model, which is trained using the UCI Adult dataset, ACGAN, FairGAN, and FairGAN+.

From Table I, created with results from source [7], we can observe that in the data generation section, the risk difference indicates a bias against females when calculated over the UCI Adult dataset. FairGAN+ shows the minimum risk difference when compared to the rest of the models. The average class-conditioned error on distribution over the pair of unprotected attributes and the protected attribute is shown by the Balanced Error Rates (BER), i.e., it measures the discrimination occurring due to the relation between the protected and unprotected attributes. From the table we can see that it is maximum in the case of FairGAN+. Hence, we can conclude that FairGAN+ works accurately and efficiently with better results in the fairness metrics for data generation, as compared to its other counterparts.

For Fair Classification, we observe that Adversarial debiasing gives a lesser difference in false positive rate in comparison to FairGAN and FairGAN+. However, FairGAN+ gives better results while comparing the risk difference and the difference in true positive rates while FairGAN shows to have a better classification accuracy overall. The Table I shows that while developing a generative model in overall comparison, FairGAN+ offers a better trade-off between data fairness as well as in classification.

IV. CONCLUSION

In conclusion, biases in AI-generated data pose significant ethical concerns, and addressing these biases is crucial to ensure fair and transparent use of the technology.

Bias in data or algorithmic biases can perpetuate and amplify existing inequalities in society, leading to discrimination against certain groups, leading to harmful consequences in various applications such as machine learning, data analysis, and decision-making processes. Therefore, fairness in generative models is an important topic that has gained increasing attention in recent years. Fairness in algorithms is essential to ensure transparent, explainable, unbiased decisions. It is of utmost importance in areas where the consequences of biased decisions can impact people's lives severely. FairGAN and FairGAN+ are two recently proposed generative models that aim to generate fair and accurate data while also promoting diversity in the generated data.

FairGAN improves upon traditional GANs by incorporating fairness constraints into the objective function, resulting in a generative model that produces fair data while also being able to generate diverse and accurate data. However, FairGAN is limited by its inability to control the degree of fairness achieved, leading to a trade-off between fairness and accuracy. FairGAN+ addresses these limitations by introducing three discriminators, each responsible for optimizing a different aspect of the generated data: accuracy, fairness, and diversity. By using multiple discriminators, FairGAN+ is able to achieve both fairness and accuracy in the generated data, while also promoting diversity. FairGAN+ has shown promising results in various applications such as image and text generation and has the potential to be applied in other areas where fairness in generated data is important.

Overall, FairGAN and FairGAN+ are promising approaches to addressing fairness in generative models. As more attention is given to the issue of fairness in machine learning and data analysis, further developments in this area are expected to lead to more sophisticated and effective techniques for generating fair and accurate data.

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